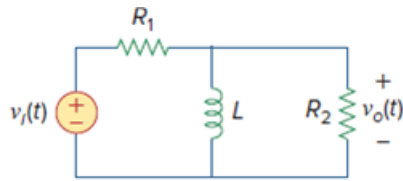


Problem

For the circuit in Fig. 14.40, obtain the transfer function $V_o(\omega)/V_i(\omega)$. Identify the type of filter the circuit represents and determine the corner frequency. Take $R_1 = 100\ \Omega = R_2$, $L = 2\text{ mH}$.

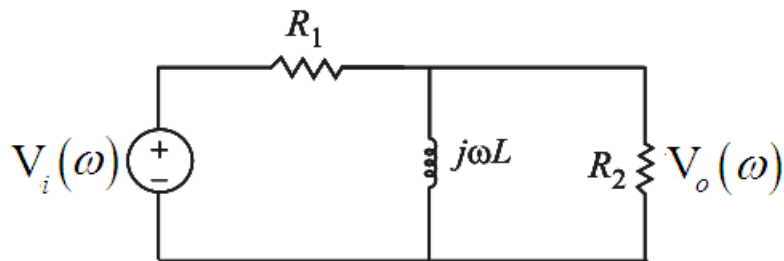
Figure 14.40:



Step-by-step solution

Step 1 of 6

The frequency domain equivalent circuit is,



Step 2 of 6

From the given circuit,

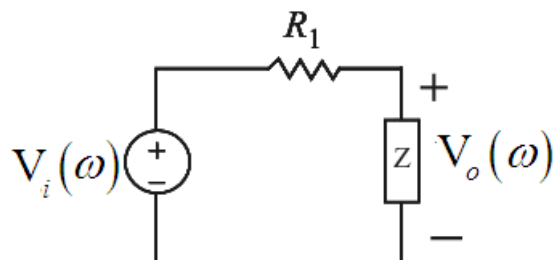
The parallel combination of R_2 resistor and $j\omega L$ impedance gives,

$$Z = j\omega L \parallel R_2$$

$$Z = \frac{j\omega L \times R_2}{j\omega L + R_2}$$

The given circuit can be simplified as

Step 3 of 6



Step 4 of 6

Using voltage division method

$$\begin{aligned}
 V_o(\omega) &= V_i(\omega) \times \left[\frac{Z}{Z + R_1} \right] \\
 \Rightarrow \frac{V_o(\omega)}{V_i(\omega)} &= \left[\frac{\frac{j\omega R_2 L}{j\omega L + R_2}}{\frac{j\omega R_2 L}{j\omega L + R_2} + R_1} \right] \\
 \Rightarrow \frac{V_o(\omega)}{V_i(\omega)} &= \frac{j\omega R_2 L}{j\omega R_2 L + j\omega R_1 L + R_1 R_2} \\
 \Rightarrow \frac{V_o(\omega)}{V_i(\omega)} &= \frac{j\omega R_2 L}{L \left(j\omega (R_1 + R_2) + \frac{R_1 R_2}{L} \right)} \\
 \Rightarrow \frac{V_o(\omega)}{V_i(\omega)} &= \frac{j\omega R_2}{j\omega (R_1 + R_2) + \frac{R_1 R_2}{L}} \\
 \Rightarrow \frac{V_o(\omega)}{V_i(\omega)} &= \frac{j\omega R_2}{(R_1 + R_2) \left(j\omega + \frac{R_1 R_2}{(R_1 + R_2)L} \right)} \\
 \Rightarrow H(\omega) &= \frac{V_o(\omega)}{V_i(\omega)} \\
 \therefore H(\omega) &= \frac{R_2}{R_1 + R_2} \left(\frac{j\omega}{j\omega + \omega_c} \right) \dots\dots\dots(1)
 \end{aligned}$$

Step 5 of 6

Where, $\omega_c = \frac{R_1 R_2}{(R_1 + R_2)L} \dots\dots\dots(2)$

From equation (1)

$$H(0) = \frac{R_2}{R_1 + R_2} \left(\frac{j}{j + \left(\frac{\omega_c}{0} \right)} \right)$$

$$H(0) = 0$$

$$H(\infty) = \frac{R_2}{R_1 + R_2} \left(\frac{j}{j + \left(\frac{\omega_c}{\infty} \right)} \right)$$

$$H(\infty) = \frac{R_2}{R_1 + R_2}$$

Step 6 of 6

Hence, given circuit is high pass filter,

Given that,

$$R_1 = R_2 = 100$$

$$L = 2 \text{ mH}$$

From equation – (2),

$$\omega_c = \frac{100 \times 100}{(100 + 100)(2 \times 10^{-3})}$$

$$\omega_c = \frac{10000}{200(2 \times 10^{-3})}$$

$$\therefore \omega_c = 25 \text{ k rad/sec}$$